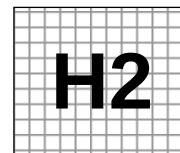


Civics Group	Index Number	Name (use BLOCK LETTERS)
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ST. ANDREW'S JUNIOR COLLEGE
2011 JC2 Preliminary Examinations

H2 BIOLOGY

9648/3

Paper 3: Applications (Mark Scheme)

Wednesday

21 September 2011

2 hours

Additional Materials: Answer Paper
 Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination,

1. Attach Question 5 to the **cover sheet** provided.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Paper 3	
1	/14
2	/13
3	/13
5	/20
Total	/60
4 (Planning)	/12

This document consists of **14** printed pages.

[Turn over

Answer **all** questions.

1.

(a) A molecular biologist is in the process of cloning the human anti-thrombin gene using bacteria cells.

Fig. 1.1 shows 2 plasmids (A and B) and the respective restriction sites of various restriction enzymes. In plasmid B, mutations were deliberately introduced and affected restriction sites are denoted by *. The lengths of the plasmid DNA between consecutive restriction sites are also indicated.

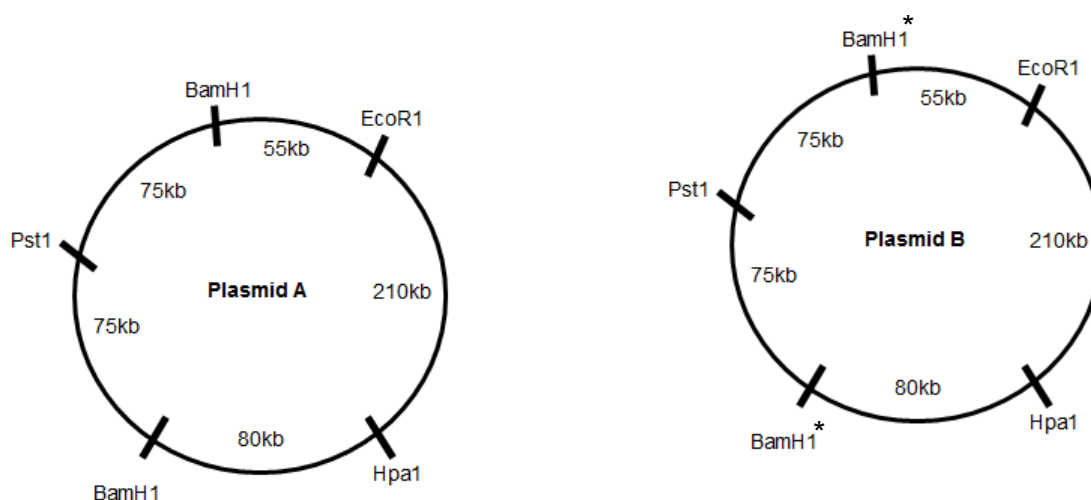


Figure 1.1

He carried out a series of 2 experiments by adding different restriction enzymes to the plasmids. The components added to each tube can be seen in Table 1.1.

Each tube was then left to incubate at 37°C for about 15 minutes (sufficient time for complete digestion to take place).

Tube number	Tube Contents
1	Plasmid A + <i>BamH1</i> + <i>Hpa1</i>
2	Plasmid B + <i>BamH1</i> + <i>EcoR1</i> + <i>Pst1</i>

Table1.1

(i) Describe the natural role of *BamH1*, *EcoR1*, *Pst1* and *Hpa1* in the bacteria in which they are produced.

.....[2]

1 To cleave; foreign DNA that enters the bacterial cell;

2 Protect bacterial cells; from bacteriophage/viruses;

REJECT cut at restriction site/gene used in genetic engineering

The DNA fragments obtained were then separated by gel electrophoresis. A DNA ladder was loaded into the left most well of the agarose gel. The biggest fragment of the ladder is 550kb and the difference between each fragment of the ladder is 50kb.

The molecular biologist then loaded the mixture from each tube into their respectively numbered wells of the agarose gel as seen in Fig. 1.2.

(ii) Predict the results of the gel electrophoresis by drawing in the bands for each of the 2 tubes in Fig. 1.2 below. Indicate the length of each band. [3]

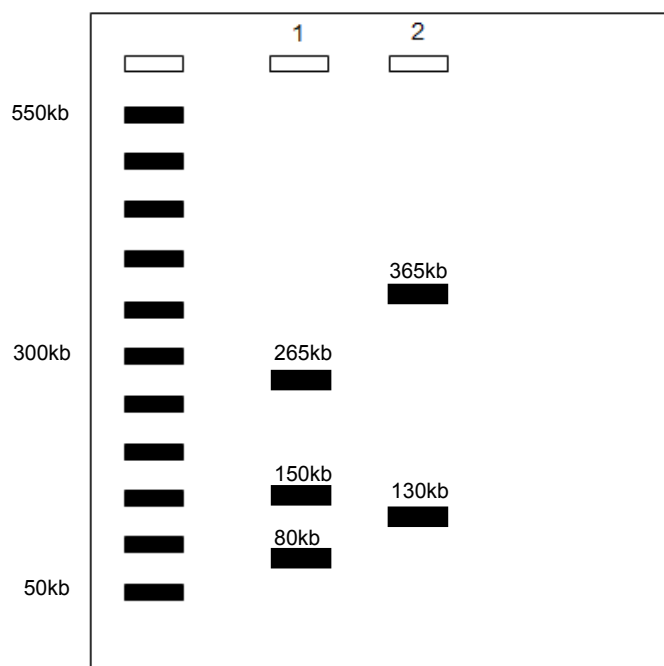


Figure 1.2

- 1 Correct position of all bands in tube 1
- 2 Correct position of all bands in tube 2
- 3 Correct length indicated of all bands in tube 1 and 2

The molecular biologist cuts plasmid A shown in Fig. 1.1 with *HpaI* in order to insert a human anti-thrombin gene.

(iii) Outline the steps that must subsequently be followed to produce a recombinant plasmid. [2]

- 1 Use *HpaI*/the same restriction enzyme; to cut the human anti-thrombin gene;
- 2 Add ligase; to form phosphodiester bonds between the ends of the cut plasmid and human gene / seal the nicks in the sugar-phosphate backbone ;

(b) Fig. 1.3 shows the results of plating the transformed bacteria cells onto an agar plate with the appropriate substances.

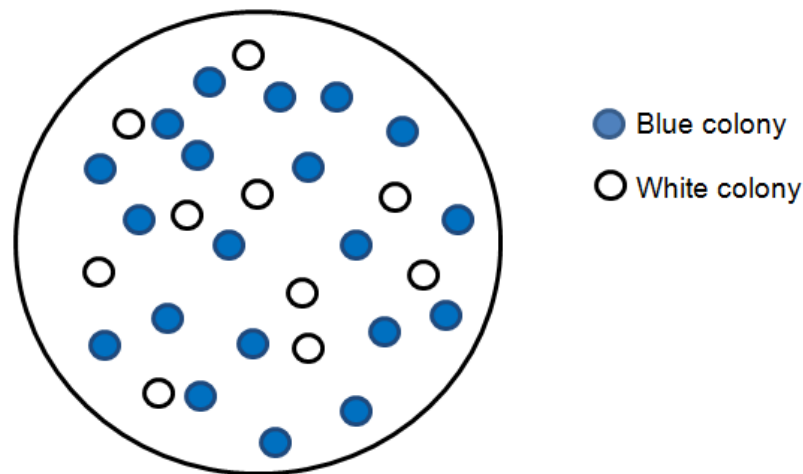


Figure 1.3

(i) Calculate the percentage of cells into which the recombinant plasmid had been successfully introduced.

.....[1]
 1 33.3%. (Working = $10/30 \times 100 = 33.3\%$)

(ii) Explain why some colonies appeared white while others appeared blue.

.....[4]
 1 lacZ gene produces functional β -galactosidase;
 2 culture medium contains X-gal; a colourless substrate that when cleaved by β -galactosidase; will turn blue;
 3 non-recombinant colonies will appear blue; with the intact *lacZ* gene produces functional β -galactosidase / synthesise functional β -galactosidase;
 4 recombinant colonies will appear white; with gene of interest inserted/disrupting *lacZ* gene / cannot synthesise functional β -galactosidase;

(c) The same molecular biologist has problems obtaining functional anti-thrombin protein in this cloning experiment. Analysis reviewed a much longer amino acid sequence in the non-functional protein. Propose a possible explanation.

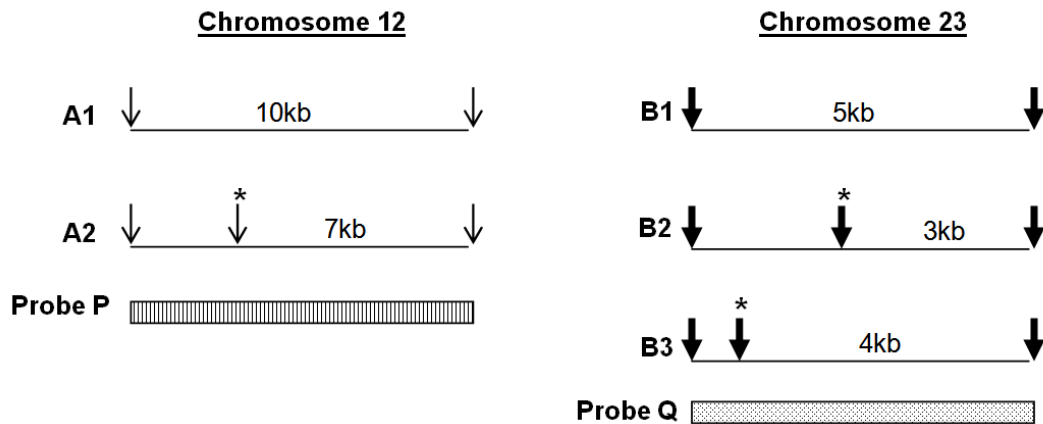
.....[2]
 1 DNA/gene containing introns was used to insert into the plasmid;;
 2 Bacteria cells lack RNA splicing machinery;;

[Q1 Total: 14]

2.

(a) The DNA of a family affected by a rare autosomal disease was analysed using RFLP. Two genes suspected to be responsible for the disease were analysed. Fig. 2.1 shows the two genes, A and B, which are found on chromosomes 12 and 23 respectively.

EcoRI was used to restrict digest locus A and two types of alleles, A1 and A2 were detected. *PstI* was used to restrict digest locus B and three types of alleles were detected. The RFLP in locus A and B were identified by probes P and Q respectively.



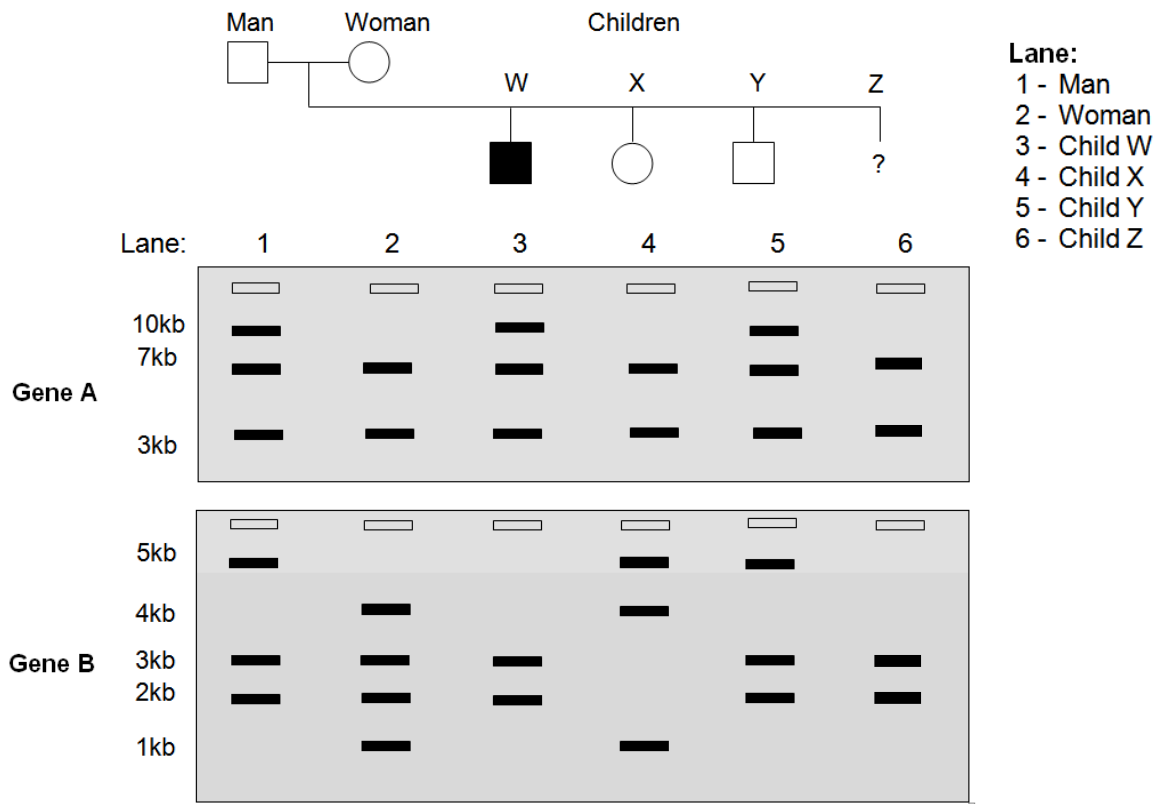
Polymorphic restriction sites were found at each locus indicated with (*).

↓ indicates *EcoRI* restriction sites.

↓ indicates *PstI* restriction sites.

Figure 2.1

Fig. 2.2 shows the pedigree chart and RFLP analysis of a family affected by the disease. A phenotypically normal couple had three children, W, X and Y, with only one of them affected by the disease.



(i) With reference to Fig. 2.2, state which gene caused the disease.

.....[1]

1 Gene B;

(ii) State the genotype of Child W and explain the inheritance of the disease.

.....[2]

1 A1A2B2B2;;

2 Child W inherited B2 allele from both parents; disease inherited in a recessive manner;

(iii) The couple recently gave birth to a fourth child, Child Z who was found to be homozygous at both gene loci. In Fig. 2.2, indicate the bands that would be obtained for Child Z in Lane 6.

.....[2]

1 Gene A : band at 7kb and 3kb;;

2 Gene B : band at 3kb and 2kb;;

RFLP can be used to detect diseases in an indirect manner.

(b) With reference to RFLP, explain the features of genetic markers that allowed them to be used for the indirect analysing of the inheritance of diseases.

.....[3]

- 1 known sequence (DNA/gene); with known location on a chromosome;
- 2 genetic marker found close to the gene of interest;;
- 3 highly polymorphic / absence or presence of restriction enzyme sites;;

(c) Other applications of RFLP are in the areas of genetic fingerprinting and in genomic mapping.

(i) Explain why genes especially protein-encoding genes are seldom used as markers for genetic fingerprinting.

.....[2]

- 1 sequence of protein-encoding genes are conserved; due to its important biological function;
- 2 mutation rate lower than non-coding region; not able to differentiate between individuals;

(ii) Explain the use of genetic markers in genomic mapping in terms of linkage mapping.

.....[3]

- 1 genetic markers are evenly spaced throughout each of the chromosomes; used to determine the order/position of genes;
- 2 frequency with which two genetic markers are inherited together; is a measure of the closeness of the two loci on a chromosome;
- 3 recombination frequency / frequency of crossing over between 2 genetic markers; indicate the distance between two loci on a chromosome in map units / centiMorgan;

[Q2 Total: 13]

3. The Cystic Fibrosis (CF) gene is found on chromosome 7 and in non-sufferers it gives rise to a protein called Cystic Fibrosis Transmembrane Conductance Regulator (CFTR). The defect which most commonly leads to the disease involves the deletion of three nucleotides from the gene. This is illustrated in Fig. 3.1.

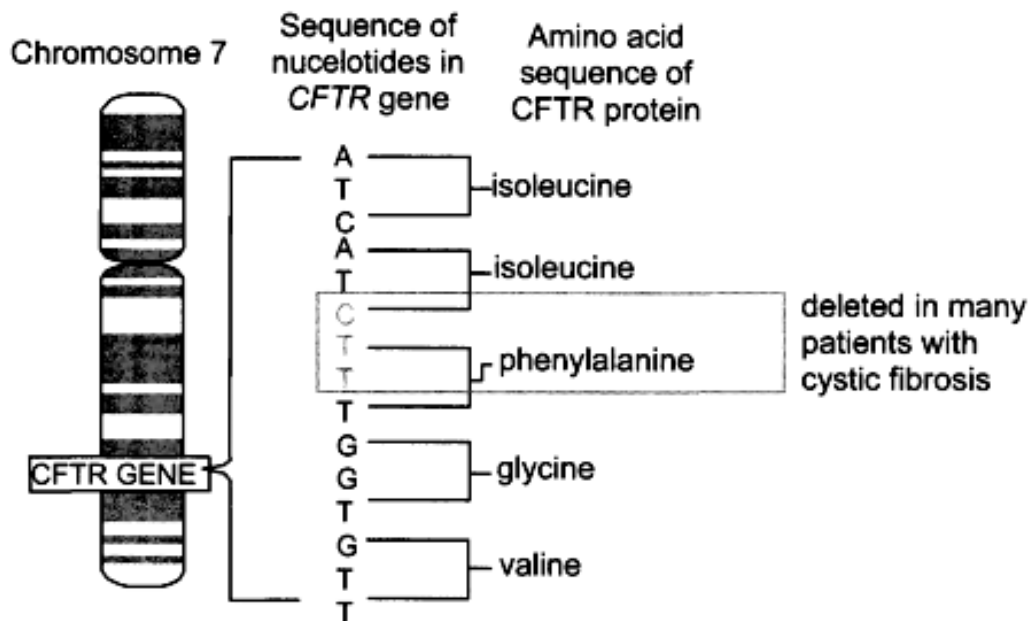


Figure 3.1

(a) With reference to Fig. 3.1, explain how the deletion of the nucleotides in the CF gene results in the production of thick and sticky mucus.

.....[4]

- 1 Deletion of 3 bases; results in loss of amino acid phenylalanine;
- 2 Change in tertiary structure of CFTR protein;
- 3 Non-functional CFTR is unable to transport Cl^- ions out of epithelial cell; accumulation of Cl^- in the cells;
- 4 causes entry and accumulation of Na^+ ions in the cell; to neutralise Cl^- ions;
- 5 Resulting in high ionic concentration/ lowered water potential in cell; and water is drawn into the cell / cannot leave the cell.

DNA coding for the normal allele was cloned and put into liposomes. The droplets were inhaled as an aerosol into the nose.

(b) Explain how a liposome is effective as a vector for introducing DNA into mammalian cells.

.....[2]

- 1 phospholipid bilayer of the liposome; mimic the structure of cell surface membranes;
- 2 entry into cell by fusion with cell membranes or endocytosis; and release transgene into cytoplasm of cell;

(c) Suggest why the *in vivo* method is used in this case and give one limitation of this approach.

.....[2]

- 1 Lung epithelial cells cannot be removed from body easily / cannot be cultured effectively in vitro in sufficient numbers / cannot be re-implanted efficiently in patients;;
- 2 Limitation 1: cannot select for successfully transfected cells
/ Limitation 2: cannot observe for insertional mutagenesis

(d) Gene therapy using the liposomes did not provide a permanent cure for CF; the trials showed that the transgene expression declined after about 2 weeks and was negligible after only 4 weeks. Suggest 2 reasons for this phenomenon.

.....[2]

- 1 transgene not integrate into the genome / remained extrachromosomal, so transgene may not segregate equally to daughter cells when cells divide / not propagated into daughter cells;;
- 2 lung epithelial cells usually non-dividing; gene not propagated when cells die;

(e) An alternative gene therapy method for CF patients is to use a modified viral vector. In a number of trials, the frequency of successful gene delivery using the viral vector was observed to decrease with repeated rounds of treatment. Suggest a reason for the decrease.

.....[1]

- 1 The introduced vectors may stimulate the immune system; to produce antibodies targeting the viral vector

(f) Explain why cystic fibrosis is a suitable candidate for treatment using gene therapy.

.....[2]

- 1 traceable to a single defective gene / mutation in 1 gene; requires just a single replacement of defective allele;
- 2 recessive disease; introduction of one normal/dominant allele produces sufficient amounts of CFTR protein; to mask effect of recessive mutation

[Q3 Total: /13]

Planning question

4. Fig. 4.1 shows an aquatic plant called *Elodea canadensis* (Canadian pondweed) which lives entirely underwater and is commonly used in aquariums.



Figure 4.1

Design an experiment using *Elodea* to test the hypothesis that **the rate of photosynthesis is dependent on the light intensity**.

Your planning must be based on the assumption that you have been provided with the following equipment and materials:

- *Elodea* (Canadian pondweed)
- sodium hydrogen carbonate
- distilled water
- table lamp
- a room which can be made dark
- sharp knife
- ruler
- plastacine
- burette
- filter funnel
- thermometer
- stopwatch
- beaker

Your plan should have a clear and helpful structure to include:

- a description of the method used including the scientific reasoning behind the method
- an explanation of the dependent and independent variables involved
- relevant, clearly labelled diagrams
- how you will record your results and ensure they are as accurate and reliable as possible
- proposed layout of results tables and graphs with clear headings and labels
- the correct use of technical and scientific terms
- relevant risks and precautions taken

Framework:

- 1** Theoretical consideration or rationale of the plan to justify the practical procedure;;
- 2** Independent variable = light intensity (distance between Elodea and lamp);
Dependent variable = volume of oxygen evolved per min;
- 3** Other variables to be kept constant - Mentions amount of sodium hydrogen carbonate used;
Monitoring of constant temperature using thermometer;
- 4** Control with details, environmental conditions;
Purpose of control;
- 5** Labelled diagram of experimental setup;;
 - a. Plasticine to hold down Elodea
 - b. Inverted burette used for measurement of gas volume
 - c. Inverted filter funnel
 - d. Ruler and table lamp
- 6** Suitable range of light intensities (at least 6 distances from the light source);
State range. Eg. 10 cm to 60 cm;
- 7** Time set aside for plant to acclimatise / equilibration procedure;
3 replicates and 3 repeats;
- 8** Tabulation of data with correct headings;;
- 9** Processing of data – light intensity;
Mean volume of gas, rate of photosynthesis;
- 10** Graph with correct axis labels and units;;
- 11** Relevant risks and precautions taken;;
- 12** Correct use of technical and scientific terms;;

Theoretical consideration

- Light is absorbed by pigments in the photosystems
/ Photoactivation of chlorophyll at the reaction centre occurs and chlorophyll becomes oxidised.
- Photolysis of water results in the production of O_2 / ref. to Hill reaction

Variables

- Independent variable = light intensity (distance between Elodea and lamp)
- Dependent variable = volume of oxygen evolved per min
- Other variables to be kept constant = CO_2 concentration in water
- Other variables to be kept constant = Temperature
- Other variables to be kept constant = Same piece of Elodea should be used each time in order to make sure that each experiment is being carried out with the same leaf surface area.

Control

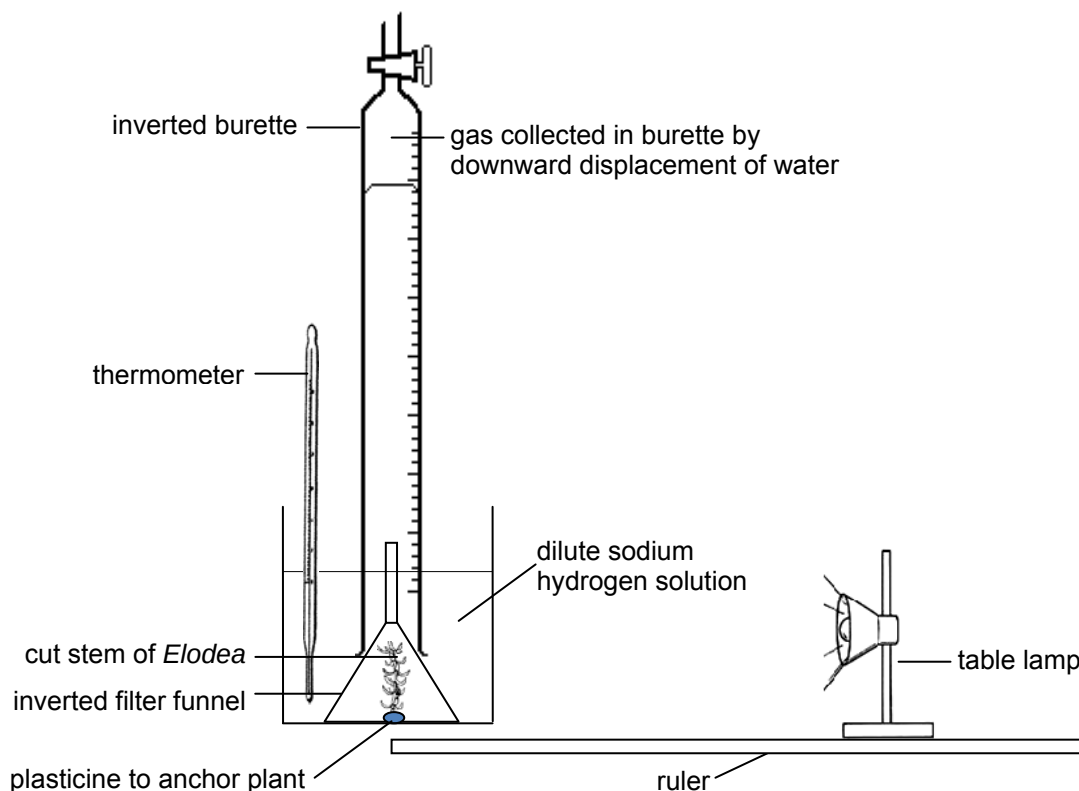
- Place the same setup in the dark
- Control being subjected to same environmental factors as that for the experiment
- Purpose of control - To calculate the amount of CO_2 evolved during respiration so as to calculate the actual amount of O_2 evolved during photosynthesis
- Rubber tubing/aquarium plastic plant (inanimate object) to replace Elodea (Reject: take away Elodea. Reason = The variable that is removed must be substituted to ensure same setup)
- Control being subjected to same environmental factors as that for the experiment
- Purpose of control - To ensure that only Elodea plant is giving out CO_2 during experiment)

Relevant risks and precautions taken

- Care with sharp knife to avoid cuts when trimming stems
- Care when handling the lamp to prevent scalding with bulb
- Dry hands when operating the lamp to prevent electric shock

Procedure

1. Add 2 g of sodium hydrogen carbonate to 600 cm³ of distilled water in a beaker.
2. Cut the stem of piece of *Elodea* to 5 cm length with a sharp knife
3. Use plasticine to attach the *Elodea* to the base of the beaker of prepared solution with the cut surface upwards.
4. Use a thermometer to monitor the temperature of the water bath at intervals throughout the experiment.
5. Set up the apparatus as below:

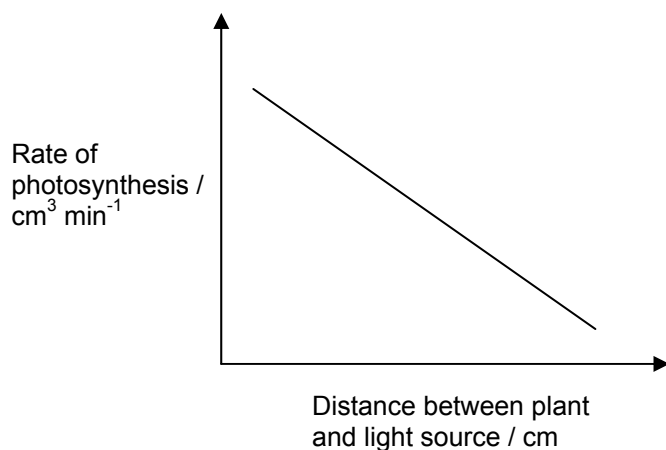


6. Darken the room. Place the table lamp 10 cm from the *Elodea*.
7. Allow the plant to adjust to the light intensity (equilibrate) for 2 min. Ensure that the rate of bubbling is constant.
8. Collect a suitable volume of gas in 5 min. Measure the volume of gas collected in the burette by downward displacement of water.
9. Repeat the procedure of increasing distances between the table lamp and *Elodea*, such as 20, 30, 40, 50 and 60 cm. In each case, allow time for the plant to equilibrate.
10. The intensity of light falling on a given object is inversely proportional to the square of the distance from the source. $\text{Light intensity} \propto 1/d^2$ where d is the distance between object and light source.

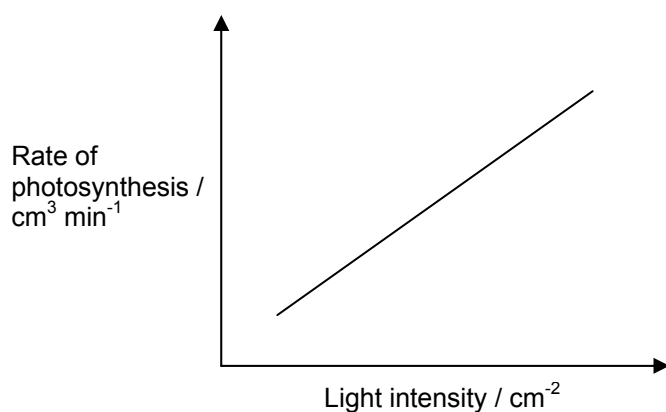
11. Tabulate your data as shown:

Distance between plant and light source / cm	Light intensity / cm^{-2}	Volume of oxygen gas evolved in 5 min / cm^3				Rate of photosynthesis (Volume of oxygen gas evolved per min) / $\text{cm}^3 \text{ min}^{-1}$
		Replicate 1	Replicate 2	Replicate 3	Mean	
10						
20						
30						
40						
50						
60						

12. Plot a graph with rate of photosynthesis on the y-axis (volume of oxygen gas evolved per min) and distance between plant and light source (cm) / rate of photosynthesis on the y-axis (volume of oxygen gas evolved per min) and light intensity on the x-axis (as $1/d^2$)



OR



- 5 Write your answers to this question on the separate answer paper provided.
Your answer:

- should be illustrated by large, clearly labeled diagrams, where appropriate.
- must be in continuous prose, where appropriate.
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- (a) Outline the process of nucleic acid hybridisation and explain how it can be used to detect and analyse restriction fragment length polymorphism. [7]**

Nucleic acid hybridisation

- 1 Restriction enzymes are used to digest DNA strands into fragments
- 2 Agarose gel electrophoresis is performed to separate the DNA fragments based on size
- 3 The gel is treated with an alkaline solution (i.e. sodium hydroxide) to cause the double-stranded DNA to denature / separate into single strands.
- 4 DNA fragments are transferred onto nitrocellulose membrane
- 5 then baked to permanently crosslink the DNA to the membrane.
- 6 membrane is treated with a single-stranded
- 7 radioactive probe which is complementary to the target sequence / marker / restriction fragment of interest
- 8 after hybridisation, excess probe is washed off from the membrane
- 9 autoradiography – pattern of hybridization is visualized on X-ray film

RFLP

- 10 restriction enzymes only cut DNA at specific restriction sites
- 11 polymorphisms / genetic variation / different alleles between individuals
- 12 difference in nucleotide sequence due to mutation
- 13 differences in the restriction sites on homologous chromosomes
/ addition or deletion of restriction sites
- 14 different lengths of restriction fragments obtained
/ different restriction fragment patterns

How to detect RFLP

- 15 digestion of genomic DNA produces numerous fragments / a smear after gel electrophoresis
- 16 radioactive probe is used to identify/distinguish the restriction fragments of interest from among many other restriction fragments

- (b) Describe the unique features of zygotic stem cells, embryonic stem cells and blood stem cells and explain their normal functions in a living organism. [8]**

Zygotic stem cells

- 1 totipotent
- 2 cells in the morula (first few divisions after fertilization / fusion of an egg and sperm)
- 3 able to differentiate into any cell type / all the tissues of the body
- 4 including extra-embryonic tissues
- 5 daughter cells are pluripotent and multipotent
- 6 Function = give rise to the whole organism

Embryonic stem cells

- 7 pluripotent
- 8 inner cell mass of the blastocyst
- 9 have the ability to differentiate into almost any cell type to form any organ or type of cell
- 10 except the extra-embryonic tissues / placenta
- 11 daughter are not totipotent but are multipotent
- 12 Function = give rise to various organs in organism / multiple specialized cell types that make up the heart, lung, skin and other tissues in the developing foetus

Blood stem cells

- 13 multipotent
- 14 able to differentiate into a limited range of cell type
/ produce only cells of a specific lineage
- 15 daughter cells are not pluripotent or totipotent
- 16 can differentiate into RBC, WBC, platelets
- 17 Function = replacement of worn out blood cells
/ daily blood turnover and fighting infections
- 18 lost through normal wear and tear, disease, injury

(c) Discuss the social and ethical considerations for the use of gene therapy.
[5]

Interfere with evolution

- 1 elimination of unwanted alleles from the gene pool
- 2 will decrease genetic variation
- 3 detrimental to the survival of our species in future
 / genes that are damaging under some conditions may be advantageous under other conditions
 / germ-line gene therapy would forever change the genetic makeup of an individual's descendants / the human gene pool would be permanently affected

Safety

- 4 question of safety and efficiency of treatment and therefore unknown long term consequences to the well-being of the recipients.
- 5 raises questions over who should participate in a human trial – the terminally ill with no treatment options or anyone who understands the potential risk but is a willing patient

Cost

- 6 gene therapy is currently very expensive, require expertise and equipment found only in major medical centres, concerns about gene therapy become a luxury available only to the rich and powerful.
- 7 possible genetic enhancements creating an advantage for those who can afford the treatment

Eugenics social policies

- 8 type of circumstances that determines whether genomes should be altered
 / what is normal or a disability or disorder, and who decides
- 9 potential for non-therapeutic enhancement possibilities / eugenic social policies
- 10 concerns about the widespread use of gene therapy making society less accepting of people who have mild disorders / genetic diseases or less able
 / lead to new definitions of “normal (e.g. intellect / height / strength etc.)

Privacy

- 11 concerns about the protection of privacy and confidentiality of medical information of patients involved in clinical trials
- 12 which may have implications on (medical) insurance coverage / employability etc.

Interfering with life

- 13 opposition to gene therapy based on religious grounds, believing that altering genetic material is against God's will / raises the specter of “playing God”
- 14 the unborn child who would be affected by germline gene therapy cannot choose whether to have the treatment

For gene therapy

- 15 Potential for tremendous patient benefit
- 16 Germline therapy can eradicate disease from subsequent generation and prevent transmission of serious genetic diseases from parent to offspring